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*Aggregate Economic Fluctuations in Endogenous Growth Models**

This paper analyzes some macroeconomic implications of introducing endogenous growth into an otherwise standard neoclassical exogenous growth model. The paper examines two types of endogenous growth models: a learning-by-doing endogenous growth model and a human-capital-investment endogenous growth model. It is found that both the learning-by-doing and the human-capital-investment endogenous growth models perform better, quantitatively, than the standard exogenous growth model in terms of explaining aggregate economic fluctuations, especially labor market fluctuations.

1. Introduction

This paper analyzes some macroeconomic implications of introducing endogenous growth into an otherwise standard neoclassical exogenous growth model.[1] By staying within the standard exogenous growth framework, we intend to isolate the contribution of assuming endogenous growth for explaining economic fluctuations. In particular, we plan to focus on the following two problems with the standard exogenous growth model as documented in Hansen and Wright (1992) and Christiano and Eichenbaum (1992). The standard exogenous growth model fails to account for the facts that hours worked fluctuate considerably more than productivity and that the contemporaneous correlation between hours worked and productivity is negative. As discussed in Hansen and Wright (1992), adding nonseparable leisure, indivisible labor, home production or government spending improves the standard exogenous growth model along the labor market dimension. This study differs from Hansen and Wright (1992) and Christiano and Eichenbaum (1992) in that the models considered there do not allow for endogenous growth.

Two types of endogenous growth models are considered. These two models differ from each other via a different human capital accumulation process in each of them. In these models, endogenous human capital ac-

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[1]By "standard exogenous growth model" we refer to models which assume exogenous economic growth due to changes in technology, competitive equilibrium with market clearing, as in Long and Plosser (1983), Hansen (1985), and King, Plosser and Rebelo (1988a).

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cumulation results from either a learning-by-doing process or investing in human and physical capital. Introducing endogenous human capital formation alters the margin governing agents' hours-worked choices. In addition, random shocks to human capital accumulation introduce a labor supply effect in the endogenous growth models which leads to a partial effect on the hours worked-productivity correlation that offsets the positive correlation due to productivity shocks leading to a negative correlation as observed in data. Following Arrow (1962) and recent work by Christiano and Eichenbaum (1988), the concept of learning-by-doing is that learning is the product of experience and takes place during activity. In a model with learning-by-doing, productivity shocks do not simply affect the marginal productivity of labor, but also affect the rate at which work effort augments human capital which in turn affects the growth rate of the economy. The second type of endogenous growth model has growth resulting from investment in human and physical capital, based on work by Uzawa (1965), and, recently, King, Plosser, and Rebelo (1988b), King and Rebelo (1990), and Rebelo (1991). In this type of model, endogenous growth arises through human capital accumulation as the outcome of deliberate actions taken by economic agents.

A number of recent papers recognize the long-run implications of endogenous growth models.[2] This paper, on the other hand, focuses on quantitative implications of endogenous growth models for economic fluctuations. Gomme (1993) considers the welfare implications of alternative monetary policies using the second type of endogenous growth model where growth results from investment in human and physical capital. In his model human capital production is not stochastic, while we allow human capital accumulation to be stochastic here.

One should note, however, that there is no formal estimation of the stochastic properties of human capital shocks. This is mainly because it is very difficult to get a handle on these shocks from actual data. As discussed in the paper, however, the stochastic properties of these shocks affect the performance of the endogenous growth models in generating U.S. business cycle facts, and may contribute to our understanding of business cycle phenomena.

To preview the results, it is found that both the learning-by-doing and the human-capital-investment endogenous growth models perform better, quantitatively, than the standard exogenous growth model in terms of ex-

[2]Recently, the long-run implications of endogenous growth models have been studied by among others, Lucas (1988), Rebelo (1991) and King and Rebelo (1990). The main focus of these papers is to be able to understand the wide cross-country heterogeneity in growth experienced within the world. It is found that, in endogenous growth framework, differences in government policies can substantially affect long-run growth rates, whereas the exogenous growth model fails in this dimension. As pointed out in King and Rebelo (1990), models of endogenous economic growth provide new analytical paths for studying old problems in the economics of development.

plaining fluctuations in many aggregate variables, especially labor market variables. The endogenous growth models match the two important labor market facts identified above.

The paper is organized as follows. Section 2 describes the exogenous and endogenous growth models, displays the choice problem of a typical household, and interprets optimality conditions. Section 3 discusses the numerical technique used to solve each model and the parameter values chosen for use in numerical simulations of the models. In Section 4, moments of simulated series are compared with moments of series drawn from actual economies to quantitatively assess each model's performance. A discussion of the quantitative results is provided in this section. The final section contains conclusions.

2. The Models

In each of the three models, it is assumed that the economy is populated by many identical agents such that each agent's influence on aggregate activities is insignificant. The representative agent attempts to maximize expected discounted utility, which depends on sequences of consumption and leisure in pure time units. The agent maximizes the utility measure:

$$\tilde{U} = E_0 \sum_{t=0}^{\infty} \beta^t \, U(C_t, L_t) \,. \tag{1}$$

Per capita values of consumption and leisure are, respectively, C_t and L_t. The momentary utility function is assumed to be strictly concave and twice continuously differentiable, E_t is an expectation operator conditional on information dated t and earlier, and $0<\beta<1$ is the subjective discount factor. It is assumed that the momentary utility function takes the form

$$U(C_t,\, L_t) = (1 - \gamma)\ln(C_t) + \gamma \ln\,(L_t) \,. \tag{2}$$

There is a single good in each of the three models considered here. The good is produced according to a neoclassical production function.

The Basic Exogenous Neoclassical Growth Model

In this model, production is assumed to take place according to the Cobb-Douglas production function

$$F(K_t,\, z_tN_t) = K_t^{\alpha}(z_tN_t)^{1-\alpha} \,, \tag{3}$$

where K_t and N_t are per capita values of physical capital and labor hours, respectively, used to produce per capita output, Y_t . z_t is an exogenous shock to labor productivity which is assumed to be the sum of a deterministic trend, μ, and random deviations, θ_t, about that trend, as in

$$z_t = \exp(\mu t + \theta_t) \; . \tag{4}$$

θ_t is assumed to evolve according to the following AR(1) representation

$$\theta_t = (1 - \rho_\theta)\theta + \rho_\theta \theta_{t-1} + \varepsilon_{\theta_t} \; , \tag{5}$$

where θ is the mean of θ_t, $|\rho_\theta| \leq 1$, and innovations ε_{θ_t} are serially independent with standard deviation σ_θ. The per capita physical capital stock evolves according to

$$K_{t+1} = (1 - \delta_k)K_t + I_t \; , \tag{6}$$

where I_t is per capita investment and δ_k is a fixed depreciation rate of capital. The representative agent faces two constraints. First, consumption plus physical capital investment cannot exceed output:

$$C_t + I_t \leq Y_t \; . \tag{7}$$

Second, the normalized time endowment is allocated between leisure and work effort

$$L_t + N_t \leq 1 \; . \tag{8}$$

Because of the nonsatiation assumption implicit in (2), one can, without loss of generality, impose strict Equality in (7) and (8).

The representative household maximizes (1) subject to (2)–(8) along with nonnegativity constraints $C_t \geq 0$, $N_t \geq 0$, $K_{t+1} \geq 0$ for $t \geq 0$, given a value for K_0. The first-order conditions for this problem are:

$$-U_{L,t} + U_{C,t}\, F_{N,t} = 0 \; , \tag{9}$$

$$-U_{C,t} + \beta E_t U_{C,t+1}\, [F_{K,t+1} + (1 - \delta_k)] = 0 \; , \tag{10}$$

where, for period t, $F_{K,t}$ is the marginal productivity of physical capital, $F_{N,t}$ is the marginal productivity of labor, $U_{L,t}$ is the marginal utility of leisure, and $U_{C,t}$ is the marginal utility of consumption. We will compare (9) and (10) with the corresponding equations from the other two models in the next section.

The Neoclassical Endogenous Growth Model through Learning-by-Doing

In this model, there are two capital stocks, physical capital, K_t, and human capital, H_t. As in the neoclassical model with exogenous growth, consumption and physical capital investment are the two uses of output so that the per capita commodity resource constraint is again given by Equation (7). Physical capital accumulation continues to take the form in (6), and the time constraint is the same as (8) where both L_t and N_t are in pure time units.

The Cobb-Douglas production function for the learning-by-doing model takes the form

$$F(K_t, z_t N_t H_t) = K_t^{\alpha} (z_t N_t H_t)^{1-\alpha} , \tag{11}$$

where H_t is the amount of human capital embodied in a typical worker as of the beginning of period t. N_tH_t is defined as effective labor units that result from N_t hours of work undertaken by an individual with H_t units of human capital. z_t is a temporary shock to the productivity of effective labor assumed to be given by

$$z_t = \exp(\theta_t) , \tag{12}$$

with θ_t evolving according to the representation in (5).

It is assumed that a worker's human capital is related to past effort as in Christiano and Eichenbaum (1988):

$$\frac{H_{t+1}}{H_t} = AN_t^{\psi_t} , \tag{13}$$

where A is a scale parameter, and ψ_t follows the autoregression

$$\psi_t = (1 - \rho_{\psi})\psi + \rho_{\psi}\psi_{t-1} + \varepsilon_{\psi_t} . \tag{14}$$

ψ is the mean of ψ_t, $|\rho_{\psi}| \leq 1$, and ε_{ψ_t} is a serially independent innovation with standard deviation σ_{ψ}. According to (13), human capital accumulation is an automatic by-product of working, to capture the idea of learning-by-doing. When $\psi_t > 0$, increased work effort accelerates human capital accumulation. Consistent with Arrow's (1962) observation that learning by experience is subject to sharply diminishing returns, the plausible values for ψ are $0 < \psi < 1$.

In this model the representative agent maximizes (1) subject to (2), (5)–(8), and (11)–(14) along with the nonnegativity constraints $C_t \geq 0$, $N_t \geq 0$, $K_{t+1} \geq 0$, $H_{t+1} \geq 0$, for $t \geq 0$, for given values of K_0 and H_0. Combining first-order conditions for this problem, the agent's choices satisfy:

$$-U_{C,t} + \beta E_t\, U_{C,t+1}\, [F_{K,t+1} + (1 - \delta_k)] = 0 , \tag{15}$$

$$U_{C,t}\, F_{NH,t}\, H_t - U_{L,t} + \beta A H_t\, \psi_t\, N_t^{\psi_t - 1}\, E_t \left\{ U_{C,t+1}\, F_{NH,t+1}\, N_{t+1} + [U_{L,t+1} - U_{C,t+1}\, F_{NH,t+1}\, H_{t+1}] \frac{N_{t+1}}{H_{t+1}\, \psi_{t+1}} \right\} = 0 , \tag{16}$$

where $U_{C,t}$, $U_{L,t}$, $F_{K,t}$ are defined earlier, and $F_{NH,t}$ is the period t marginal productivity of effective labor units.

Equation (15), governing physical capital accumulation, is exactly the same as Equation (10) in the previous model.

Equation (16) governs the decision for working. To identify the relevant margins, suppose that the amount of time spent in the production activity goes up by one unit and simultaneously leisure is reduced by one unit of time. As a result of an additional unit of time devoted to production, effective labor units increase by the amount of human capital H_t, which leads to an increase in output of $F_{NH,t}$ units. The utility value of this additional output is the first term in Equation (16). The second term is the cost in utility terms of decreasing leisure by one unit. In this model, in contrast to the exogenous growth model, the amount of time spent in the production activity in period t has an effect on consumption and leisure in period $t + 1$ through human capital accumulation. In the current model, human capital accumulation is an automatic by-product of working and a one-unit increase in the amount of time spent in period t in the production activity increases the end-of-period human capital H_{t+1} by $AH_t\psi_t N_t^{\psi_t - 1}$ units. This allows the household additional consumption in period $t + 1$, the utility value of which is given by the first term in the braces. At the same time, the increase in H_{t+1} decreases the amount of time spent in the production activity in period $t + 1$ by $(N_{t+1}/H_{t+1}\psi_{t+1})$ units. This is because the household accumulates skill through learning-by-doing in period t and will, for a given value of H_{t+2}, work fewer hours in period $t + 1$. This in turn means additional leisure for the household in period $t + 1$, the utility value of which is given by the first term in the square bracket. The negative effect of having additional leisure is less consumable output in period $t + 1$, the utility cost of which is reflected in the second term in the square bracket in (16). As a whole, the term in braces gives the discounted expected utility value of increasing the end-of-period human capital stock due to a one unit increase in the amount of time spent in production at time t.

The Neoclassical Endogenous Growth Model through Human Capital Investment

As in the previous model there are two types of capital stocks, physical capital, K_t, and human capital, H_t, and a single good. In contrast to the previous endogenous growth model, however, human capital or an individual's skill level is not simply the by-product of work. In the current model, the way in which an individual allocates physical capital and time in the current period affects the individual's human capital level in future periods. Final output production takes place according to a Cobb-Douglas production function. If N_t is the fraction of pure time units allocated to the production of final output and ϕ_t is the fraction of a period's physical capital stock allocated to final output production, the final output production function can be written as

$$F^m(\phi_t K_t, z_t N_t H_t) = A_m(\phi_t K_t)^\alpha (z_t N_t H_t)^{1-\alpha}, \tag{17}$$

where A_m is a scale parameter, $N_t H_t$ represents effective labor units, and z_t is the same productivity shock as in the previous endogenous growth model.

The remaining physical capital stock, $(1 - \phi_t)K_t$, and time not allocated to final goods production or leisure, $D_t = 1 - N_t - L_t$, are used in human capital production. It is assumed that human capital production takes place according to the following constant returns to scale technology to ensure that a constant rate of human capital accumulation is feasible (see King, Plosser, and Rebelo 1988b)

$$F^h((1 - \phi_t)K_t, z_t^h D_t H_t) = A_h[(1 - \phi_t)K_t]^\sigma (z_t^h D_t H_t)^{1-\sigma}, \tag{18}$$

where A_h is a scale parameter, and $D_t H_t$ represents effective labor units used in human capital production. z_t^h is a temporary shock to the productivity of effective labor in human capital production assumed to be given by

$$z_t^h = \exp(\theta_t^h), \tag{19}$$

where θ_t^h evolves according to

$$\theta_t^h = (1 - \rho_{\theta^h})\theta^h + \rho_{\theta^h}\theta_{t-1}^h + \varepsilon_{\theta_t^h}. \tag{20}$$

θ^h is the mean of θ_t^h, $|\rho_{\theta^h}| \le 1$, and $\varepsilon_{\theta_t^h}$ is a serially independent innovation with standard deviation σ_{θ^h}. The evolution of the stock of human capital is governed by:

$$H_{t+1} = F^h((1 - \phi_t)K_t, z_t^h D_t H_t) + (1 - \delta_h)H_t, \tag{21}$$

where δ_h is a fixed rate of depreciation on human capital.

The physical capital stock accumulates as in the previous models according to (6). Since consumption and investment are the two uses of final output, the commodity resource constraint is the same as in Equation (7).

The representative agent maximizes (1) subject to (2), (5)–(7), (12) and (17)–(21) along with nonnegativity constraints $C_t \ge 0$, $N_t \ge 0$, $D_t \ge 0$, $\phi_t \ge 0$, $K_{t+1} \ge 0$, $H_{t+1} \ge 0$, for $t \ge 0$, and given values of K_0 and H_0. The first-order conditions for this problem are:

$$-U_{L,t} + U_{C,t} H_t F_{NH,t}^m = 0, \tag{22}$$

$$-U_{L,t} + \beta H_t F_{DH,t}^h E_t \left\{ N_{t+1} F_{NH,t+1}^m U_{C,t+1} + \left[\frac{D_{t+1}}{H_{t+1}} + \frac{(1 - \delta_h)}{H_{t+1} F_{DH,t+1}^h} \right] U_{L,t+1} \right\} = 0, \tag{23}$$

$$K_t F^m_{\phi K,t} U_{C,t} - \beta K_t F^h_{(1-\phi)K,t} E_t \left\{ N_{t+1} F^m_{NH,t+1} U_{C,t+1} + \left[\frac{D_{t+1}}{H_{t+1}} + \frac{(1-\delta_h)}{H_{t+1} F^h_{DH,t+1}} \right] U_{L,t+1} \right\} = 0 , \tag{24}$$

$$-U_{C,t} + \beta E_t U_{C,t+1} \left[\phi_{t+1} F^m_{\phi K,t+1} + (1-\delta_k) \right] + \beta E_t (1-\phi_{t+1}) \frac{F^h_{(1-\phi)K,t+1}}{H_{t+1} F^h_{DH,t+1}} U_{L,t+1} = 0 , \tag{25}$$

where $F^m_{\phi K,t}$ and $F^m_{NH,t}$ are, respectively, the period t marginal productivities of physical capital and effective labor units in final output production. $F^h_{(1-\phi)K,t}$ and $F^h_{DH,t}$ are, respectively, the period t marginal productivities of physical capital and effective labor units in human capital production.

Equation (22) governs the decision for working in final output production. If the allocation of time to final output production N_t goes up by one unit, effective labor units increase by the amount of human capital H_t, which leads to an increase in final output production by $F^m_{NH,t}$ units, the utility value of which is the second term in (22). The first term is the cost in utility terms of decreasing leisure by one unit.

Equation (23) governs the decision for working in human capital production. If the allocation of time to human capital production D_t goes up by one unit, the household has one less unit of leisure, the utility cost of which is the first term in Equation (23). The return to increased current human capital production is the increase in end-of-period human capital of $H_t F^h_{DH,t}$ units, which allows the household additional consumption in period $t + 1$, the utility value of which is the first term in the braces. The decrease in the allocation of time to human capital production next period, due to an increase in end-of-period human capital, is given by $[(D_{t+1}/H_{t+1}) + (1 - \delta_h)/H_{t+1}F^h_{DH,t+1}]$, the utility value of which is the second term in the braces. As a whole the term in braces gives the discounted expected utility value of increasing end-of-period human capital by $H_t F^h_{DH,t}$ units.

Equation (24) governs the decision for allocating physical capital across the two sectors. If the fraction of physical capital used in final output production goes up by one unit, output increases by $K_t F^m_{\phi K,t}$ units, which allows the household additional consumption in period t, the utility value of which is the first term in Equation (24). The effect of this decision is to decrease end-of-period human capital by $K_t F^h_{(1-\phi)K,t}$ units which means fewer units of consumption in period $t + 1$, the utility cost of which is the first term in the braces. The second term is the utility cost of increasing the allocation of time to human capital production in period $t+1$ due to a decrease in end-of-period human capital production. The term in braces as a whole gives the discounted expected utility cost of decreasing end-of-period human capital by $K_t F^h_{(1-\phi)K,t}$ units.

Equation (25) is associated with the investment decision. To identify the margins, suppose that a unit of the good is allocated away from period t consumption, toward physical capital accumulation with the end-of-period human capital stock H_{t+1} held fixed. The unit increase in end-of-period physical capital K_{t+1} provides a net return in final output production in the next period given by $[\phi_{t+1} F^m_{\phi K,t+1} + (1 - \delta_k)]$. This allows the household additional consumption in period $t+1$, the discounted expected utility value of which is given by the second term in Equation (25). Given the unit K_{t+1} increase, for H_{t+1} to be unaffected the allocation of time to human capital production in period $t+1$ will be decreased by $(1 - \phi_{t+1})(F^h_{(1-\phi)K,t+1}/H_{t+1} F^h_{DH,t+1})$. The discounted expected utility value of the extra leisure afforded by the decreased time allocated to human capital formation is given by the third term in Equation (25). The first term in Equation (25) is the utility cost of allocating a unit of the good away from period t consumption, toward investment.

Rearranging first-order conditions (22)–(25), one can show that

$$\frac{F^m_{\phi K,t}}{F^m_{NH,t}} = \frac{F^h_{(1-\phi)K,t}}{F^h_{DH,t}}, \tag{26}$$

which is a familiar efficiency condition in production. According to (26), the marginal rate of transformation between physical capital and effective labor units must be equated across the two sectors.

3. Solution Procedure and Parameter Values

Numerical Solution of the Models

Since the models are highly nonlinear, it is not possible to find closed-form solutions. We solve the models using a linear-quadratic approximation procedure which involves: (i) linearization of the optimality conditions, with equilibrium conditions imposed, by taking a first-order Taylor series expansion about the models' nonstochastic steady-state; (ii) conjecturing decision rules for choice variables that are linear in the state variables and shocks; (iii) determining coefficient values for the linear decision rules using a method of undetermined coefficients. Details of the procedure used are in Christiano (1990).

Since the procedure involves approximating the efficiency conditions around a stationary point, it requires that the variables display no growth. In the exogenous growth model, z_t is the engine of growth. Hence, variables in the exogenous growth model are rendered stationary by dividing all variables in the system that grow by the growth component, $\exp(\mu t)$.

In the endogenous growth models, the productivity shock z_t is stationary and makes no contribution to growth. In the nonstochastic steady state of these models, growth in output, consumption, the capital stock and investment, equals the growth in H_t. Hence, the transformation of the endogenous growth models involves dividing these variables by H_t.

Parameter Values

Baseline parameter values for preferences, technologies, and the shock processes for all three models are given in Table 1. The share of leisure in the utility function, γ, is set to 0.76. This value, given the other parameter values, conforms to the sample average of the leisure-market activity ratio for U.S. economy.[3] The discount factor, β, is set to 0.99264 so that the steady-state real interest rate is 1% per quarter.[4] On the technology side, the labor share parameter, $(1 - \alpha)$, is set to 0.65 which conforms with the postwar data for the U.S. economy.

The depreciation rate of physical capital is set to 10% per annum or 0.0252 per quarter. In the exogenous growth model μ is set equal to 0.0041. This value conforms with the average quarterly growth rate of per capita U.S. GNP over the period of 1970:*i*–1993:*iv*. In the learning-by-doing model, the nonstochastic steady-state value of ψ is set to a value $\psi = 0.00069$ such that the model generates a steady-state growth rate of per capita GNP equal to the steady-state growth rate of the exogenous growth model. Following King and Rebelo (1990), in the two-sector endogenous growth model the constant term A_m in final output production is normalized to one, and the constant term in human capital production sector A_h is set to a value such that the model generates a steady-state growth rate equal to the rate of growth for the other two models.

Assessing plausible values for the parameters of the human capital production function and human capital depreciation in the two-sector endogenous growth model is difficult. This is due to the fact that the human capital good is composed of various activities which result in an improvement in labor efficiency such as education, health, construction and maintenance of public goods, etc. However, there has been little research on the parameters of individual technologies for investment in human capital (see Rosen 1976 and Heckman 1976). In this paper it is assumed that these different

[3]The sample average of leisure-market activity ratio for the U.S. economy, 1970:*i*–1993:*iv*, is 3.69. In the steady-state calculations, increasing leisure share in the utility function drives the steady-state leisure-market activity ratio up.

[4]A quarterly real rate of 1% is approximately the average rate of return on capital for the sample period (see Schlagenhauf and Wrase 1995).

activities can be put together in one sector referred to as the education sector or schooling. The labor share parameter, $(1 - \sigma)$, in the human capital production function is set equal to 0.95, which makes human capital production relatively more labor intensive compared to final output production. This is roughly in accordance with Uzawa (1965) and Lucas (1988) who assume that only labor is used to produce human capital. There is a variety of evidence on the magnitude of the depreciation of human capital (see Mincer 1974 and Haley 1976). In the baseline simulations, δ_h is set to 0.0252.[5]

In all three models, the parameters of the stochastic process that governs the technological shock are based on estimates by Burnside, Eichenbaum, and Rebelo (1993), who estimate a first-order autoregression for the linearly detrended logarithm of the technology shock, z_t. Based on this estimate, ρ_θ is set to 0.9857 and σ_θ is set to 0.01369. The mean of Θ_t is assumed to be one. There is not much evidence on the parameters of the stochastic processes for the human capital in the two endogenous growth models. Following the stochastic process for the technological shock, the mean of Θ_t^h is assumed to be one, and the AR(1) coefficients, ρ_ψ and ρ_{θ^h}, are set equal to 0.9857.[6] The standard deviations of the shock terms, σ_ψ and σ_{θ^h}, are chosen such that the standard deviation of the simulated output series is close to the standard deviation of U.S. output over the period 1970:*i*–1993:*iv*. Thus, σ_ψ is set to 0.000087 and σ_{θ^h} is set to 0.007. It is assumed that there is no correlation between σ_θ and σ_ψ in the learning-by-doing model and between σ_θ and σ_{θ^h} in the human-capital-investment model although implications of non-zero cross-correlations are discussed in the discussion section.

4. Quantitative Results

This section investigates quantitative properties of each model described above. For comparability, data drawn from actual U.S. time series, 1970:*i*–1993:*iv*, and data obtained from simulating the model economies have been first logged and then detrended by the Hodrick-Prescott filter. The statistics reported below for the model economies are generated performing Monte Carlo simulations using 100 iterations for 96 periods each, which is equal to the number of quarters in the sample data. The shock terms are drawn from a random number generator with unconditional mean values of the shock terms chosen as starting values.

[5]The sample average of capital-output ratio for U.S. data, 1970:*i*–1993:*iv*, is 10.30. Increasing capital shares in the human capital production drives the steady state capital-output ratio up, and decreasing the human capital depreciation rate increases the steady-state capital-output ratio relative to baseline results. However, the simulation results reported below are not sensitive to the parameter values for σ and δ_h.

[6]It is also the case that the simulation results reported below are not sensitive to the AR(1) coefficients, ρ_ψ and ρ_{θ^h}.

TABLE 1. *Baseline Parameter Values*

Parameter	Value	Description
β	0.99264	*subjective discount factor*
γ	0.76	*leisure share in utility*
α	0.35	*capital share in output production*
σ	0.05	*capital share in human capital production*
δ_k	0.0252	*physical capital depreciation rate*
δ_h	0.0252	*human capital depreciation rate*
θ	1	*mean value of productivity shock*
ψ	0.00069	*mean value of shock term to learning-by-doing*
θ^h	1	*mean value of shock term to human capital production*
A	1.005	*scale factor in learning-by-doing*
A_m	1	*scale factor in output production*
A_h	0.02349	*scale factor in human capital production*
μ	0.0041	*steady state exogenous growth rate*
ρ_θ	0.9857	
σ_θ	0.01369	
ρ_ψ	0.9857	
σ_ψ	0.000087	*parameters of shock processes*
ρ_{θ^h}	0.9857	
σ_{θ^h}	0.007	

Cyclical Implications of Each Model

Table 2 reports some second moment properties of each model along with comparable U.S. data moments for the sample period 1970:*i*–1993:*iv*. For each variable listed in the table, the first row provides information on moments for the U.S. data. The business cycle facts apparent in the data rows in the table are that output is about two times as volatile as consumption and investment is about three times as volatile as output. Hours are slightly less variable than output. All variables are positively correlated with output. In terms of labor market phenomena, hours fluctuate more than productivity and the contemporaneous correlation between hours and productivity is slightly negative.[7]

In the exogenous growth model, the volatility of each variable relative to output is lower than what is found in the U.S. data. The volatility of

[7]Hansen and Wright (1992) report that depending on the hours series and the sample period, hours are slightly less variable than or about as variable as output, but fluctuate more than productivity, and the correlation between hours and productivity is near zero or slightly negative.

TABLE 2. *Moments for Select Variables*

Variable v_t	Models	Standard Deviation: Percent	Standard Deviation: Relative to Output	Correlation of Real Output with V_t
Output	*Data*	1.76	1	1
	Exog	1.58	1	1
	LBD	1.76	1	1
	HCI	1.76	1	1
Consumption	*Data*	0.92	0.52	0.68
	Exog	0.69	0.43	0.95
	LBD	0.77	0.43	0.93
	HCI	0.65	0.37	0.78
Investment	*Data*	6.32	3.59	0.9
	Exog	4.14	2.63	0.99
	LBD	4.59	2.6	0.98
	HCI	4.02	2.29	0.98
Hours	*Data*	1.56	0.89	0.66
Worked	*Exog*	0.74	0.47	0.98
	LBD	1.6	0.91	0.8
	HCI	1.81	1.02	0.78
Productivity	*Data*	1.38	0.78	0.53
	Exog	0.86	0.55	0.99
	LBD	1.05	0.6	0.43
	HCI	1.17	0.67	0.28

Variable	Models	Hours worked versus productivity: Relative Standard Deviation	Hours worked versus productivity: Contemporaneous Cross Correlation
Productivity	*Data*	1.13	−0.29
	Exog	0.86	0.94
	LBD	1.52	−0.17
	HCI	1.55	−0.35

NOTE: Exog is exogenous growth model, LBD is learning-by-doing endogenous growth model, and HCI is human-capital investment endogenous growth model.

The models' artificial data have been first logged and then detrended using the Hodrick-Prescott filter. The standard deviations and correlations computed from the models' artificial data are the sample means of statistics computed for each of 100 simulations. Each simulation has 96 periods, the number of quarters in the U.S. data.

The U.S. data are quarterly and are divided by population, logged, and detrended using the Hodrick-Prescott filter. All variables are expressed in constant 1987 dollars. Output is measured by gross domestic product, consumption by consumption of nondurables and services, investment by gross fixed investment, and hours by civilian employment. Productivity is defined by output divided by hours.

Source of data: The OECD's Quarterly National Accounts. The sample period is 1970:*i*–1993:*iv*.

consumption series relative to output is slightly low, but for work effort and productivity, the model generated data do not fluctuate enough relative to output. The relative volatility of investment is 2.63 in the model and 3.59 in the U.S. data. In general these results are similar to the results obtained from the standard exogenous growth models in Hansen (1985), King, Plosser, and Rebelo (1988a) and Hansen and Wright (1992). The model fails to match the volatility of work effort relative to productivity and does poorly in terms of matching cross correlation properties of hours and productivity data with output. Finally, the model predicts that hours worked and productivity are highly correlated, 0.94, but in the data the contemporaneous cross correlation of the two variables is −0.29.

The second moment properties of the endogenous growth model through learning-by-doing improve on those of the exogenous growth model especially in terms of labor market implications. The volatility of work effort relative of output is higher than what the exogenous growth model predicts, and close to the U.S. sample counterpart. The volatility of productivity relative to output is still less than the U.S. data but higher than the exogenous growth model. The volatility of hours worked relative to productivity is 1.52 in the model and 1.13 in the data. The model improves the contemporaneous cross correlation of hours worked with output, and the contemporaneous cross correlation of productivity with output compare to the exogenous growth model. At the same time, the model is able to predict the negative correlation between productivity and hours. In terms of the other variables, the model is able to predict the volatility of consumption relative to output closely, but, as in the exogenous growth model, the volatility of investment relative to output is lower than the data.

In the endogenous growth model through human capital accumulation, the relative volatility of consumption is slightly less variable than the exogenous growth model, and the volatility of investment is low compared to the U.S. sample counterpart. The relative volatility of hours worked and productivity increases compared to the exogenous growth model. The model is able to increase the volatility of hours worked relative to productivity. Finally, the model is able to predict a negative contemporaneous cross correlation between hours worked and productivity: −0.35 versus 0.94 in the exogenous growth model.

In summary, both the learning-by-doing and the human-capital-investment endogenous growth models perform quantitatively better than the standard exogenous growth model on several dimensions, especially the labor market dimensions. As detailed in the next section, the reason why the endogenous growth models considered in the paper perform better than the standard exogenous growth model in terms of labor market implications becomes clear when one considers the effects of learning-by-doing and human-capital-investment specification on agents' hours worked choices and the dynamic responses of work effort and productivity to each shock.

Discussion

As argued in Hansen and Wright (1992), one of the deficiencies of the standard exogenous growth model is that the model fails to match the volatility of hours worked relative to productivity which indicates that agents in the model are simply not sufficiently willing to substitute leisure in one period for leisure in other periods. Nonseparable preferences as in Kydland and Prescott (1982) and indivisible labor as in Hansen (1985) have been suggested to correct the model along this dimension. Both nonseparable leisure and indivisible labor models generate a more elastic short-run labor supply as compared to the exogenous growth model, hence for a given productivity shock hours fluctuate more than productivity. As detailed in Equation (16), in the learning-by-doing endogenous growth model, on the other hand, the learning-by-doing specification alters the margin governing agents' hours-worked choices. In the model, human capital accumulation is a by-product of working, hence a positive productivity shock augments human capital and gives agents a stronger incentive to substitute leisure in one period for leisure in other periods since the higher is the rate of return to working the greater is the cost of current leisure. Then, agents will be more willing to substitute intertemporally for a given productivity shock. This in turn increases the short-run labor supply elasticity in the model and leads to a high volatility in hours worked relative to productivity.

As the optimality conditions (22)–(26) indicate, the human-capital-investment endogenous growth model, on the other hand, in addition to intertemporal substitution effects, allows for intratemporal substitution between the two sectors at a given point in time. A positive productivity shock in the final output production sector leads to an incentive for agents to substitute between the two activities at a point in time to eliminate productivity differentials across the two sectors. In this model, by working less in the human capital production sector, agents can increase hours worked in the output sector without reducing leisure as much, which in turn increases the short-run labor supply elasticity of the model.

The other deficiency of the standard exogenous growth model is that the model fails to predict the negative contemporaneous cross correlation between hours worked and productivity. In the standard exogenous growth model, the productivity shock is the only force driving the system which can be interpreted as shifting the labor demand along a stable labor supply curve and generating a positive relationship between hours worked and productivity. As argued in Christiano and Eichenbaum (1992), and Hansen and Wright (1992), one way to overcome the counterfactually high positive correlation between hours worked and productivity is to introduce economic impulses into the standard exogenous growth model which shifts the labor supply function. To explain why the endogenous growth models perform better than the exogenous growth model in this respect, one needs to look at the dynamic responses of hours worked and productivity to each shock in the endogenous growth models. In Figures 1 and 2, it is assumed that a period is one quarter and shocks take place in quarter 4.

Figure 1 displays the adjustment path of hours worked and productivity toward steady state after a one-time, one-standard deviation productivity shock in each model. To see the dynamic responses of variables due solely to a productivity shock, the standard deviation of the shock term to human capital accumulation is set equal to zero in endogenous growth models. It can be seen in Figure 1 that a positive productivity shock leads to a contemporaneous increase in hours worked and productivity in all three models. Intuitively, a positive productivity shock leads to a stochastic shift in the marginal product of labor thereby increasing the labor demand. The labor demand shifts along the labor supply curve producing a positive relationship between hours and productivity.

Figure 2 displays responses of hours worked and productivity to only a one-time, one-standard deviation shock to human capital accumulation in each endogenous growth model. Intuitively, a positive shock to learning-by-doing augments human capital or an individual's skill level and thereby increases the rate of return to working for any given current wage rate. This induces agents to substitute leisure for work since the higher is the rate of return to working the greater is the cost of current leisure. Figure 2 shows that due to this labor supply effect a positive learning-by-doing shock leads to a decrease in productivity in the current period. A positive shock to human capital production in the other endogenous growth model, on the other hand, leads to an incentive for agents to substitute final output for human capital production at a point in time to eliminate productivity differentials across the two sectors. In this model, by working less in the final output production sector, agents can increase hours worked in the human capital production. Due to this intratemporal substitution effect, as Figure 2 shows, hours worked decreases and productivity increases in the final output production sector. Both of these stochastic shocks can be interpreted as shifting the labor supply curve along the labor demand curve, creating a labor supply effect in the models. When both shocks are present—the first effect producing a positive relationship between hours and productivity, the second effect producing a negative relationship—the endogenous growth models considered in the paper are able to predict the negative contemporaneous correlation between hours and productivity.

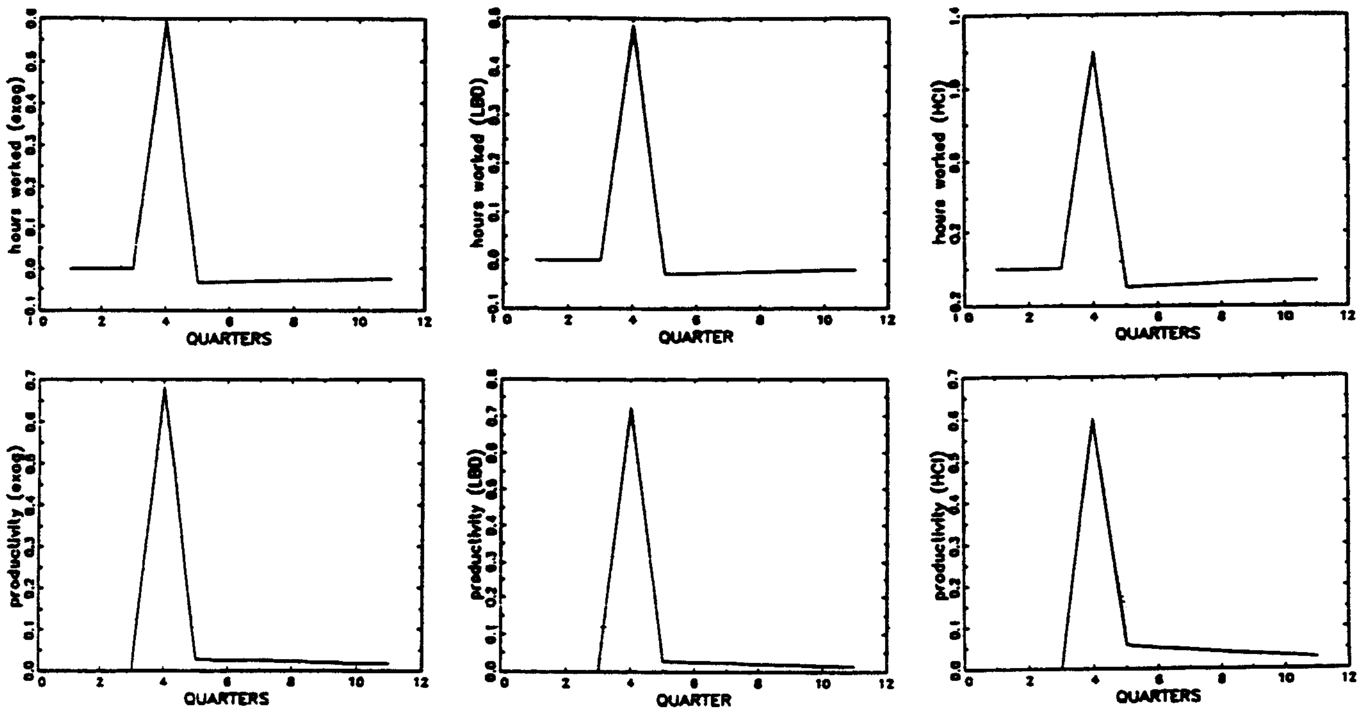

Figure 1.
Impulse responses in terms of percentage deviation from steady state to a one-standard-deviation shock to productivity in period 4. Productivity is defined as output divided by hours.

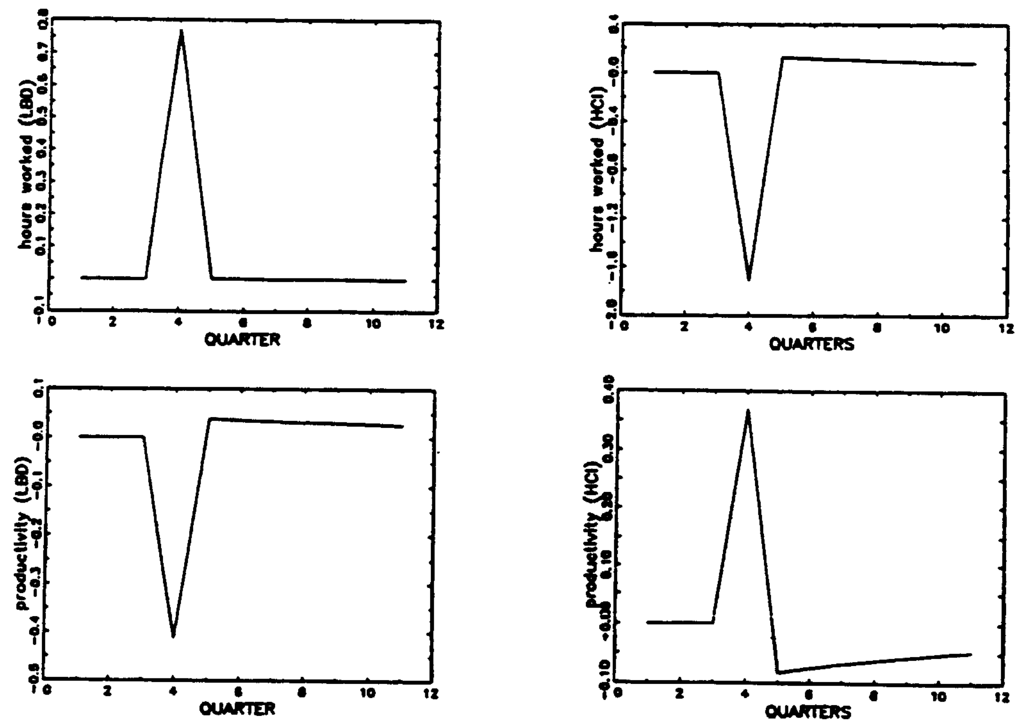

Figure 2.
Impulse responses in terms of percentage deviation from steady state to a one-standard-deviation shock to human capital accumulation in endogenous growth models in period 4.

One should note that the labor market implications of the endogenous growth models are sensitive to the stochastic properties of the human capital shocks. Unfortunately, there is little evidence to guide one in choosing the standard deviation of the shocks, as well as the cross-correlation between the productivity and human capital shocks. In both models, as the standard deviation of human capital shocks increases, the labor supply effect gets stronger. This in turn affects the cross correlation between hours and productivity. If σ_ψ is increased to 0.000095 in the learning-by-doing model, the correlation between hours and productivity becomes -0.23. An increase of σ_{θ^h} to 0.008 in the human-capital-investment model leads to a correlation of -0.42. With more variable human capital shocks, the standard deviation of output is 1.81 and the volatility of hours relative to productivity is 1.60 in the learning-by-doing model. They are 1.92 and 1.74, respectively, in the human-capital-investment model. In the simulation analysis above, the standard deviation of the stochastic shocks are set to values such that the volatility of output series from the models matches the volatility of output series in the U.S. data.

Non-zero cross correlations have different implications in the two models. In the learning-by-doing model, since human capital accumulation is a by-product of working, a positive cross correlation between the shocks increases agents' willingness to substitute leisure in one period for leisure in other periods. This results in a more elastic labor supply compared to the zero correlation case and increases the volatility of hours relative to productivity. When the cross correlation between σ_θ and σ_ψ is set to 0.10, the volatility of hours worked relative to productivity is 3.85 and the volatility of output is 2.741. In the human-capital-investment model, the cross correlation between σ_θ and σ_{θ^h} affects agents' incentive to move across the two sectors. Lower values indicate that the technology and human capital shocks more frequently take on different values across sectors which in turn implies a greater incentive to move resources across sectors and increases the volatility of hours relative to productivity in the model. When the cross correlation between σ_θ and σ_{θ^h} is increased to 0.10, the volatility of hours relative to productivity decreases to 1.27, and the volatility of output decreases to 1.65.

5. Conclusions

This paper has examined the implications of endogenous growth for economic fluctuations and especially for labor market activity. Two types of endogenous growth models have been considered: a model of endogenous growth through learning-by-doing and a model of endogenous growth through human capital investment. The paper shows that the endogenous growth models perform quantitatively better than the standard exogenous growth model. It is also found that the theory predicts that the stochastic properties of the human capital shocks affect the ability of the models to generate the business cycle facts which should stimulate researchers to think more seriously about these human capital accumulation technologies and their stochastic properties.

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